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Integration of MPD and Direct Pressure Measurements to Provide Risk informed Pore Pressure Ranges for Process Safety in Challenging Infill Drilling in Differentially Depleted HPHT Formations of N. Kuwait Jurassic Gas

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Abstract

The North Kuwait Jurassic Gas (NKJG) carbonate reservoirs are of middle Jurassic age and comprise six main units (or flow zones) comprising carbonate limestones, dolomites and evaporites.

These are HPHT (high pressure, high temperature) and critically sour tight carbonate reservoirs with pervious dolomite and very tight/ impervious limestone matrix and anhydrite vertical flow barriers. These were originally overpressured reservoirs, all the flow zones in these dolomite and limestone units were on a narrow pressure band (11000 to 12000 psi, or 14-15 ppg equivalent mud weight) in the pre production stage. Due to varied depths and productivity, different flow zones were put on development at different places of the field as part of field development strategy. Magnitude of differential pressure across different flow zones is magnified with increased production, resulting in drilling complications leading to stuck pipe, bore hole stability issues.

A good understanding of robust uncertainty range of expected formation pressure is critical for safe design of drilling, as well as safe execution of infill drilling which can meet business objectives of desired production levels. This operational excellence was achieved by integrating the indirect pressures (from drilling) with direct pressure measurements and reworking the formation pressure uncertainty range as an input for well and drilling design.

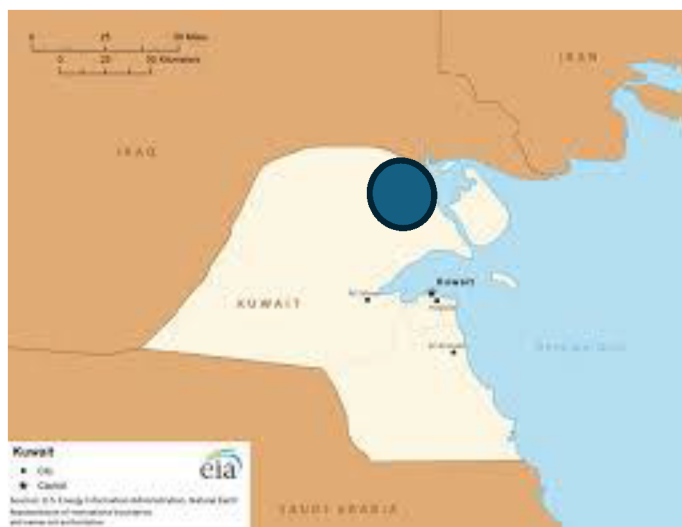


Figure 1—Location map of North Kuwait Jurassic Gas (NKJG) Fields

Pore pressure prediction is crucial in drilling because accurate estimation allows for optimized drilling fluid density and well design, directly preventing catastrophic events like blowouts and wellbore collapse. Without it, there could be immense risks including loss of well control, formation damage, significant non-productive time, and severe damage to equipment and personnel. Precise pore pressure knowledge helps define safe drilling windows, enables effective well completion, and aids in mapping hydrocarbon migration paths.

Below schematic shows the workflow for achieving the desired outcome for well delivery.

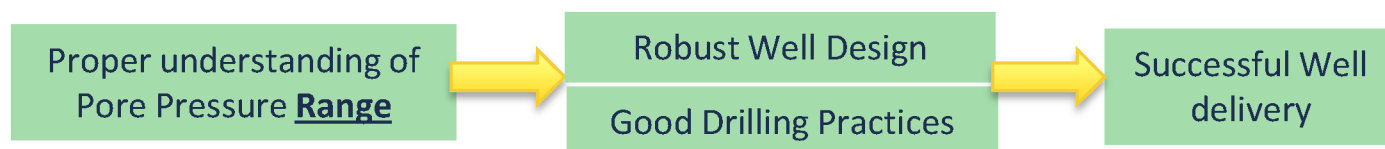


Figure 2—Pore pressure prediction and impact on well delivery

However, a single point pore pressure prediction is most of the times impossible as well as can lead to inadequate understanding of possible ranges. In order to fully appreciate the uncertainty range, the best approach is scenario thinking and coming up the full pressure range and corresponding probability of each of the ranges.

A customer centric and integrated approach is necessary to make sure there is common understanding and alignment on how the pressure ranges have arrived and what scenarios could lead these ranges.

Integration of pressure available from different sources (both direct and indirect pressure measurement during drilling operations) and deploying the scenario based pore pressure ranges, helped in achieving a maneuverable drilling window which can lead a successful well delivery with reduced drilling complications. This process safety improvement resulted in operational excellence in challenging infill drilling of NKJG wells and helped in disciplined delivery of field development plan (FDP).

Five-point pore pressure prediction was deployed which covers all the potential scenarios leading to different pore pressure outcomes. Therefore, the pore pressure ranges were worked out which included the design range (reasonable low case – reference case -reasonable high case). This range allows for designing the well and mud weight planning for safe and optimum operations. At the same time Minimum pressure and Maximum pressure are used as safety range to ensure there is adequate response to safely overcome these scenarios in case the ranges have materialized.

The full ranges of PPP from 5-point methodology are shown below:

Min Pressure	Reasonable Low case	Ref. case-	Reasonable high case	Max case
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Introduction

The North Kuwait Jurassic Gas (NKJG) carbonate reservoirs folded and fractured structure. These are over-pressured carbonates, at more than 15,000 ft, beneath Gotnia Salt and interbedded with anhydrites and kerogen-rich shales (source rock) which act as seals.

The PVT properties vary both laterally and vertically across the NKJG fields, with an overall increase in volatile components towards the north east of the NKJG area. The oil density varies 40-50 API and GOR varies from 2000 to 5000 scf / stb. The insitu reservoir fluid is critically sour with H₂S and CO₂ content carrying from 1000 to 100000 ppm.

As the key value driver of field development is gas recovery and oil or condensate recovery are considered as a byproduct, the FDP strategy is to have depletion production of different flowzones, this would mean that pressure maintenance option is dropped, and reservoir pressure is expected to drop with the production. The saturation pressure of different reservoir fluid varies from 3000 to 5000 psi, it is expected that reservoir pressure decline will get gentler as the reservoir pressure falls below that saturation point due to solution gas evolution. Same behaviour is observed in field and reservoirs where current reservoir pressure has fallen below the saturation point, leading to increase in gas oil ratio (GOR).

A schematic overview showing reservoir complexity, depth, reservoir fluid type is shown in figure below.

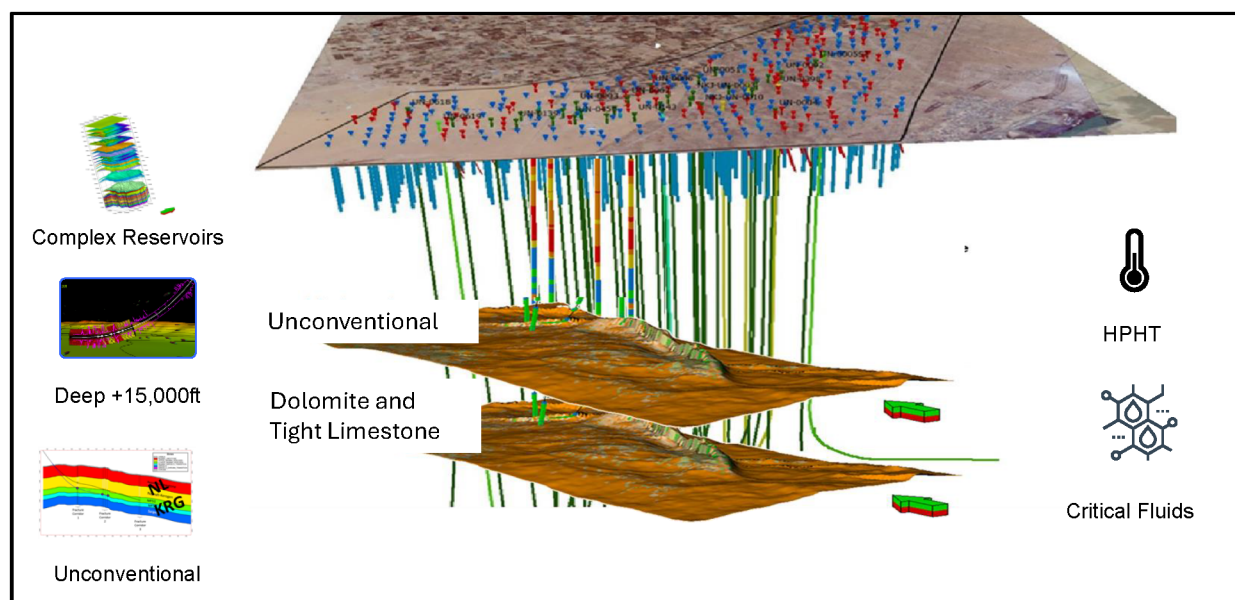


Figure 3—Schematic showing NKJG reservoir overview

The main carbonate reservoir is divided into 8 flow zones to effectively develop, monitor the reservoir. As can be seen in below figure, with increasing production over time, some flowzones have significantly depleted which others are still close to initial pressure.

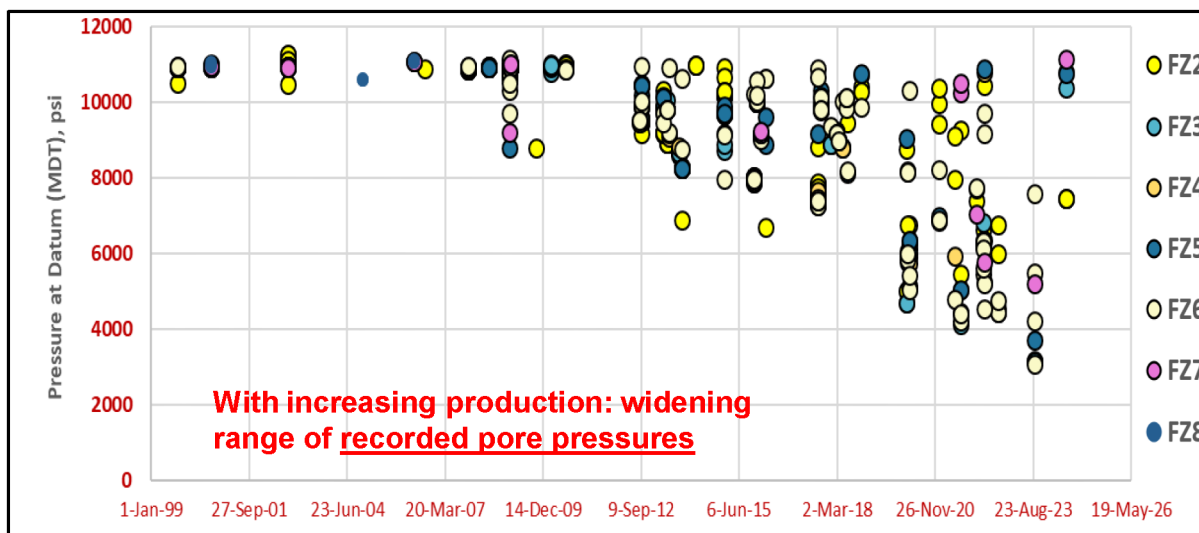


Figure 4—Differential pressure depletion over time

A typical composite log of a well is shown in figure below. It can be seen that different flowzones comprise both dolomitic and tight limestone formation. Though there is no explicit vertical barrier, but light limestone may act a vertical flow barrier causing the differential pressure depletion between the dolomite sandwiched between the limestone. On an average 20 – 30% rock is dolomitic while remaining 70 – 80% is mainly tight limestone. Though limestone has little to no storage (in place) potential and is not a target development reservoir, but infill drilling has to pass through this tight part. The tight limestone does not have flow potential on its own, but natural fractures present in this tight part may still be preserved at high pressure and might have flow potential, causing significant challenges during the drilling.



The HPHT carbonate reservoirs in North Kuwait have lithostatic initial pressure of ~11500 psi, however due to perforation and completion in different flow zones, current pressures in different flowzones varies from 2500 to upto initial pressure (~11500 psi), this would translate into differential pressure range varies upto ~8000 psi between different flowzones, posing significant challenge to well and drilling planning. Recent drilling suffered from this challenge resulting in several unplanned sub-optimal holes and sidetracks caused by mechanical issues during drilling.

The direct pressure measurements, including shut in pressure survey (SPG), pressure build up survey (BUP), formation pressure measurements (MDT or XPT) mostly indicate the permeable dolomite formations which are main target of development and production. However, the dolomitic rock is sandwiched between significantly tighter matrix of limestone where direct pressure measurements attempt like MDT or XPT will not work. At the same time acquiring formation pressure measurements MDT/XPT can cause operation challenges apart from being cost intensive.

Managed pressure drilling (MPD) was increasingly deployed to provide flexibility and somewhat larger drilling window in these depleted zones. Managed Pressure Drilling (MPD) is a drilling technique that precisely controls wellbore pressure to improve drilling efficiency and safety, particularly in challenging environments. It involves managing the annular pressure profile in the wellbore by regulating downhole pressure conditions and the hydraulic pressure around the annulus. MPD helps prevent well control problems like kicks and lost circulation, enabling drilling in narrow pressure windows and reducing non-productive time.

An empirical estimates of formation pressure of tighter zones was made using drilling equivalent circulation density (ECD) during connecting gas events or minor kick events.

Source of formation pressures

Direct Pressure Measurement

Direct pressure measurements include shut in bottom hole pressures (SBHP), pressure build up surveys (BUP) in completed wells; while in open holes, the MDT or XPT formation pressure measurements are carried out.

Though direct measurements are more accurate measurements, but SBHP and BUP surveys cannot tell the different pressures regimes when more than one flowzones are completed during the production. Similarly, this pressure will reflect on the pressure of permeable dolomite part, and will not provide any information on the tight limestone.

Some benefits of MDT or XPT formation pressure measurements in open hole are listed in below table

Table 1—XPT/MDT Formation Pressure Measurements' strengths and limitations

STRENGTH	LIMITATION
Precise measurement of formation pressure.	Scarce, operationally challenging to acquire.
Ability to record both low and high pressures, confirm significant differential depletion across and within the flow-zones.	Data acquisition in only 1/3rd of MM reservoir (deliverable reservoir: dolomite and dolomitic limestone)
	Cannot measure pressure in tighter formations, 2/3rd of MM (non-dolomitic tighter formations)

Indirect Formation Pressure Inferences

Traditional pore pressure estimation approach involves the use of normal compaction trendline (usually on resistivity, density or acoustic slowness data) to replicate the porosity loss trend with depth. However, such approach is applicable to shales only. The compaction trendline approach fails in carbonates. If such a method is utilized, at best, it may only give a local, empirical fit, and the next well drilled may exhibit a very different pore pressure profile. One of the main reasons that traditional pore pressure prediction fails in carbonates is because the rocks are characterized by a porosity variation controlled not only by effective stress but also a wide variety of physical parameters, such as variable depositional, dissolution, and diagenetic fabrics. The drilling exponent-based pore pressure estimation methods, as well, bears very high uncertainties due to lithological complexities and intricacies owing to drilling and bit technologies. Formation fluid influx event (i.e., kick), if experienced during drilling, provides a direct calibration of the carbonate pore pressure. In absence of such influx events, connection gas/pump-off gas data can be used as an indirect proxy of formation pore pressure, where the increasing or decreasing trend of the same is translated to the decreasing or increasing downhole mud overbalance, respectively.

To address these limitations, an integrated workflow was developed that couples the connection gas (CG) measurements with downhole equivalent circulating density (ECD) within a reservoir-engineering framework. Drill-cutting X-ray diffraction (XRD) analyses discriminate tight limestone facies from permeable dolomites. During MPD connections, CG flow rates and corresponding ECD values are recorded. Assuming a linear Darcy flow relationship between gas flow and pressure underbalance (ΔP = difference between formation pore pressure and drilling fluid pressure), paired CG-ECD data yield interval-specific underbalance magnitudes. Calibration against downhole wireline pressure measurements in a reference dolomite layer, assumed to be in equilibrium, establishes a bespoke CG-pressure correlation for each flow zone.

The result is a formation pressure vs. connection-gas model that provides pore pressure estimates based on CG occurrences, where downhole measurements are unavailable.

Suitable candidate wells from each field were identified for this analysis. Suitability was primarily determined based on the availability of minimum required data set. Careful consideration was also given to the well location and their distribution, to achieve a fair coverage and representation across the field. We utilized the ECD data from managed pressure drilling (MPD) that maintains a predefined downhole ECD level by applying surface backpressure during connections. Each connection gas event was mapped to the corresponding MPD ECD. First, the calibration factors are derived within nonreservoir limestone intervals, based on the availability of the direct pressure measurements in the closest overburden dolomites, assuming a pressure equilibrium. Such calibration exercise was performed at each flow zone level, focusing on the non-reservoir tight limestone facies which usually exhibit connection gases when mud weight is reduced. Once the calibration factor is quantified for each flow zone as well as each part of the field, we apply the model in adjacent exercise wells lacking direct measurements to constrain limestone pressure inferences from reported connection gases.

Figure 6 presents an example from one of the candidate wells that presets the direct and indirect pressure data from the Middle Marrat interval. Track 1 shows the gamma ray, representing an entirely carbonate interval. Track 2 and 3 present the dolomite and calcite % based on the XRD analysis of the drill cuttings, indicating limestone dominated interval. The direct measurements by XPT tool (Track 4) depicts the wide pressure distribution even within the reservoir facies. Data shows higher reservoir pressure above XX415 ft and below XX760 ft, while the middle interval (XX415-XX760 ft) hosts the more depleted dolomite facies. However, the gas data shows repeated connection gas (CG) occurrences in this middle interval, clearly indicating certain degree of unbalance against the non-reservoir limestone facies. Adopting the new workflow, based on the established CG-ECD calibration parameters, we estimated the limestone pressure, as presented the red cross marks (on Track 4) against the CG occurrences. For this studied well case, the indirect pressure inferences in the limestone facies are in pressure equilibrium with the lesser depleted dolomite facies observed above XX415 ft.

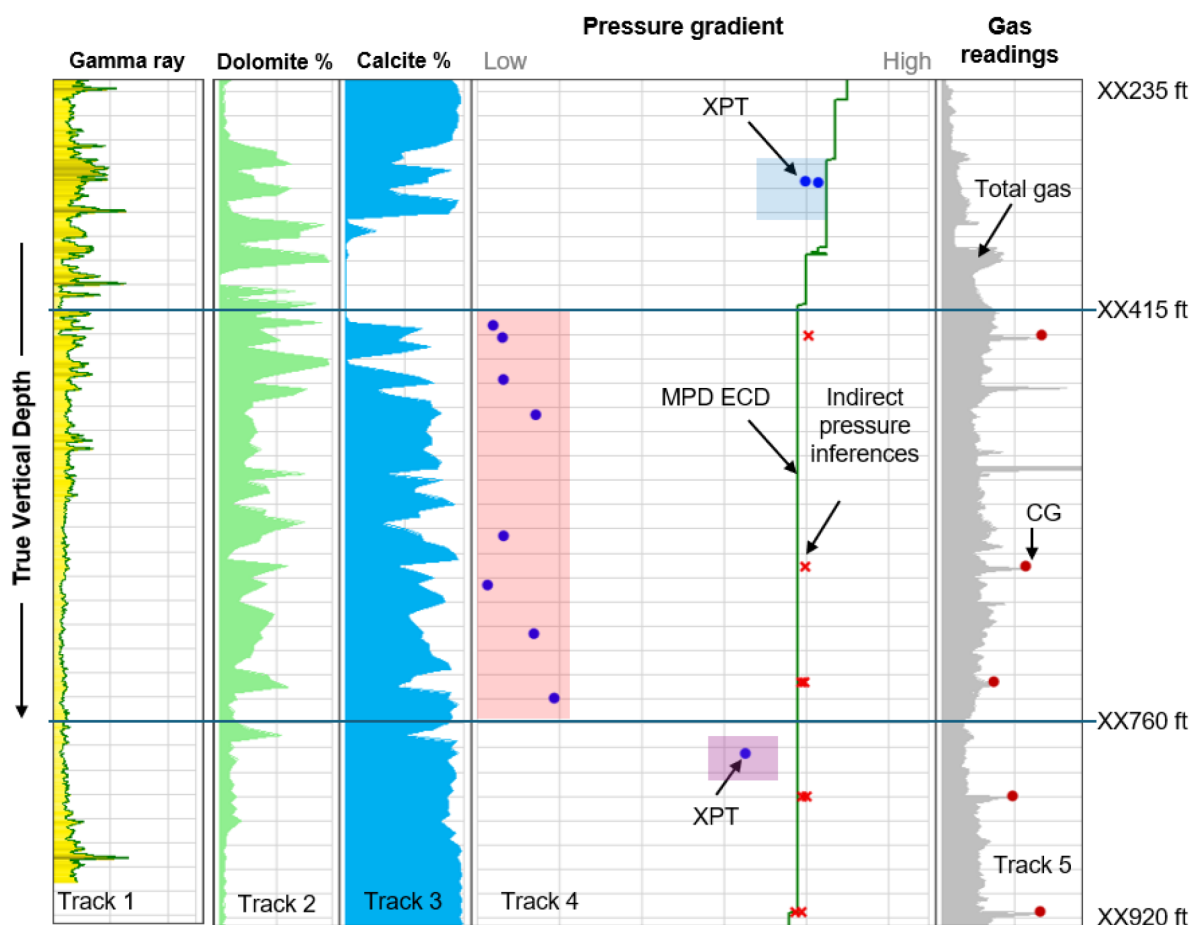


Figure 6—Example from a candidate well with direct formation pressure measurements and indirect pressure inferences based on the new workflow utilizing connection gas (CG) events. The vertical axis denotes true vertical depth (TVD). All tracks, from left to right, present data scaled from minimum to maximum values.

Based on the multi-well field scale exercise, the inferred pressure data were compiled for each field section, in all studied fields. One of the field-sector models is presented in Figure 7. The data shows a wide pressure data distribution within reservoir facies, while the indirect pressure inferences against the non-reservoir facies, based on the new workflow, shows a consistent narrow range of pressure distribution. The XRD calcite % against these indirect pressure data points reveal dominantly > 75 %, confirming dominantly non-reservoir limestone facies.

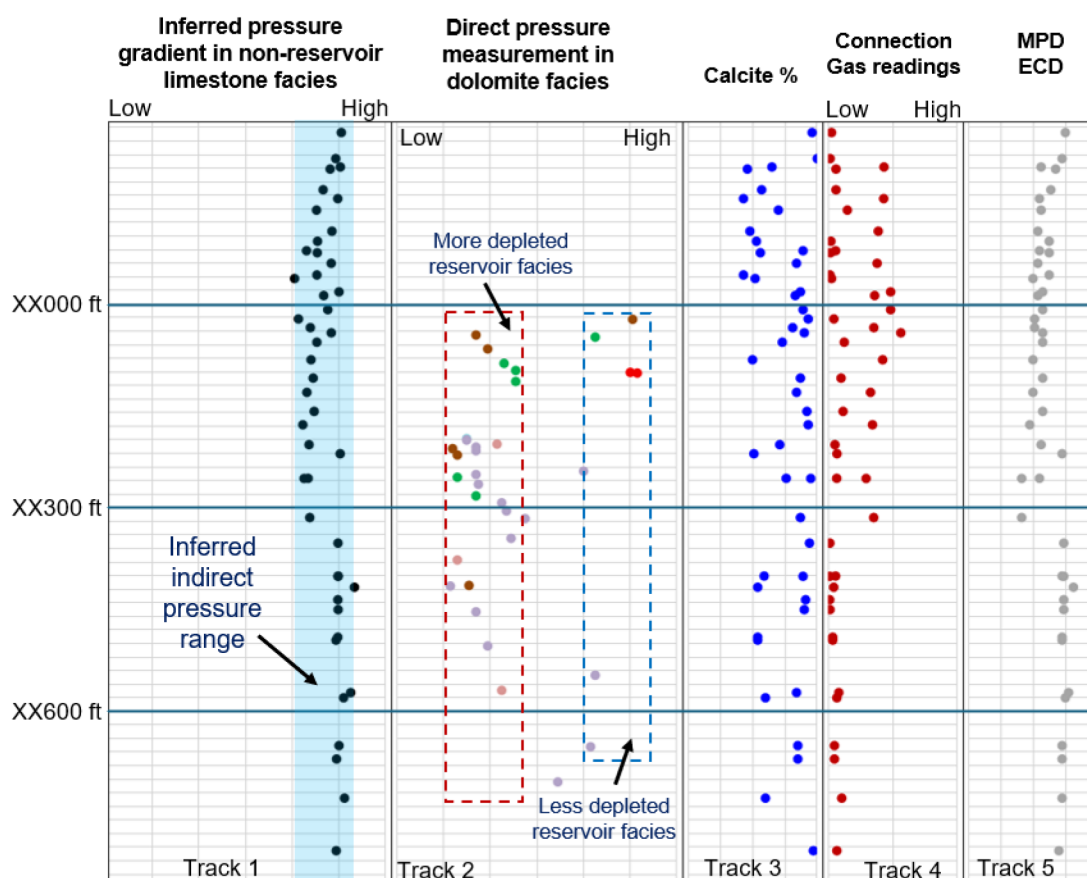


Figure 7—Field-sector pressure profile: direct formation pressure measurements (Track 2) versus CG-based indirect inferences (Track 1) via the new workflow. All tracks, from left to right, present data scaled from minimum to maximum values.

Lessons learned and implications for improved drilling and well delivery

The novel methodology was applied to the major producing fields which satisfied relevant data requirements for the workflow. It yielded an inferred indirect formation pressure gradient ranging between 0.55-0.70 psi/ft in tight limestones across various Middle Marrat flow zones. This inferred pressure ranges were further validated with the formation fluid influx events (kill mud weight). The pressure interpretation from the influx events corroborates well with the indirect pressure inferences that used the coaction gas-based new workflow. This provided further confidence and validation of the newly adopted workflow to constrain formation pressure in the tight non-reservoir limestone facies. Whereas the reservoir facies display broad pressure variability due to variable magnitude of depletion, our new workflow provided a narrow, consistent pressure envelope—demonstrating the workflow's robustness and its capacity to constrain carbonate pore-pressure in data-sparse settings.

Understanding the pressure regimes in carbonates is critical for safe drilling. A too low or too high mud weight can result into losses or kick respectively, and loss of wellbore in extreme situations. Integrating the reference limestone pore pressure into pre-section geomechanical assessment reduced uncertainty in mud weight design and provided informed- optimization opportunities in subsequent wells. Application of this approach demonstrated improved wellbore stability with informed mud weight optimization, aiding to successful well delivery without compromising subsurface objectives. Geomechanical model calibration incorporating the new pressure data refined effective stress estimates, helping to predict fracture initiation and collapse pressures more accurately. Overall, the method delivered a robust, cost-effective pore pressure estimation in tight, non-reservoir carbonates, directly informing drilling strategies and enhancing operational safety and efficiency.

Integration of Pressure data

Integration of pressure data from various sources is a key step in the workflow of generating robust pore pressure ranges.

Table below shows the pore pressure range in 5 point methodology and a transparent description of the basis of the ranges. As bulk of the carbonate to be drilled is of limestone nature, the reference case, high case and max case are governed by indirect pressure measurements. However main dolomite reservoir is only 20-30% of the total carbonate to be drilled and this is generally depleted, so the direct pressure measurements are the basis for low case estimate in the overall pressure range. It is worthwhile to note that geomechanics team evaluate the fracture gradient at the depleted pressures to make sure it is factored in the drilling window estimation.

Table 2—Transparent description of basis of PPP scenario selection

Basis of PPP ranges

Min	Min pressure for dolomitic formation including drawdown
Low	Pressure for dolomitic formation from direct measurement
Ref	Expected Pressure for non-dolomitic formations from indirect measurement
High	Likely high Pressure for non-dolomitic formations from indirect measurement
Max	Highest virgin pressure in the field

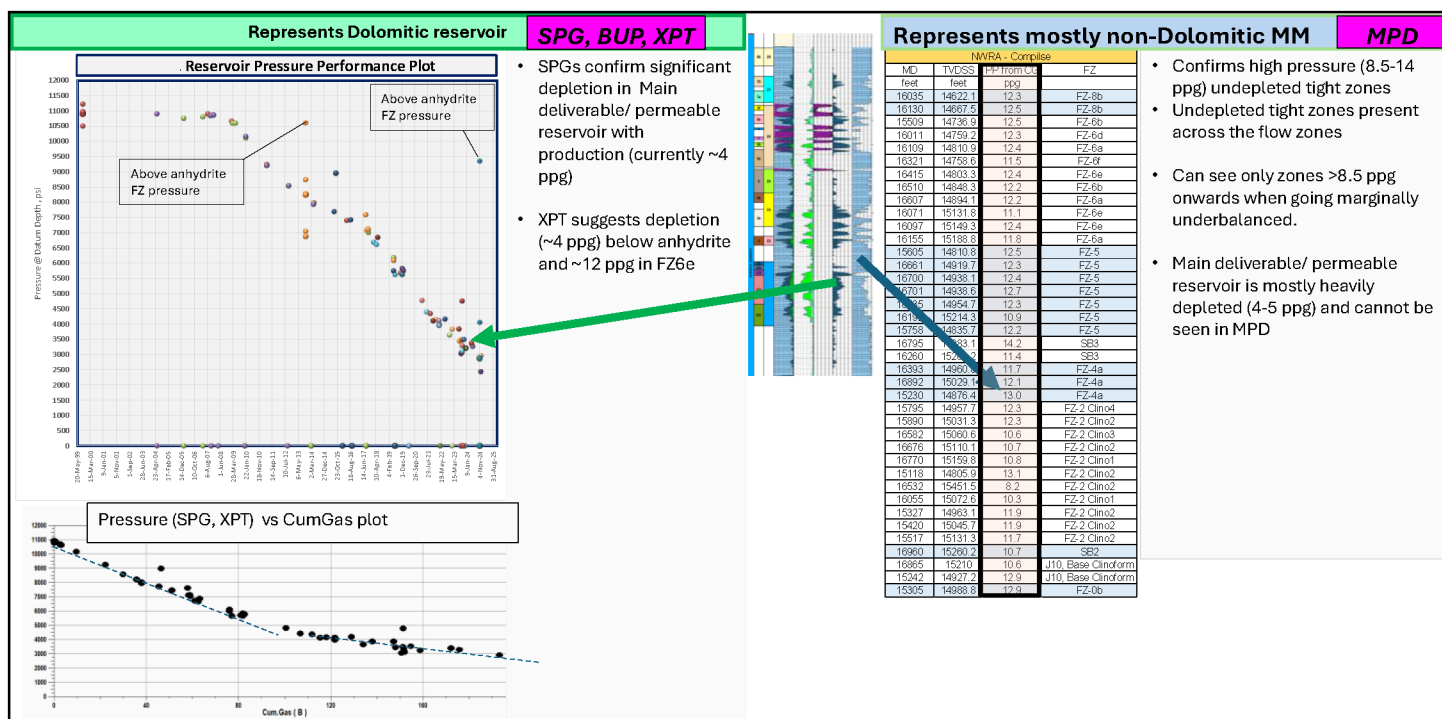


Figure 8—Pressure data integration example (both dolomitic and non-dolomitic)

Results, Observations, Conclusions

Integration of direct pressure (primarily representing the previous dolomite) and indirect pressure from drilling (associated with fractures in tight limestone matrix) allowed to construct a pressure uncertainty

range (low-mid-high) which was wide enough to capture the uncertainty level however at the same time it was maneuverable from drilling standpoint to deliver a safe and desired infill well in differentially depleted formations.

The cross plot of connection gas with MPD ECD helped to build better understanding of the relationship. Caving fingerprinting during drilling operations helped to reduce the depth uncertainty of potentially failing zones. Connection gas during MPD and fingerprinting of cavings helped to confirm presence of much less depleted tight zones which are otherwise not seen during production behaviour, but important during drilling (70% of total reservoir section)

Realized benefits by incorporating the indirect pressure measurement in the drilling mud weight design and ECD workflow and related efficiency in the operation, are listed below:

- Average time to drill reservoir section in new wells improved
- Overall reduction in NPT
- Fewer 3 7/8" holes despite increased differential depletion
- Disciplined delivery of Production gain (by mitigating the sub-optimal hole and completion)

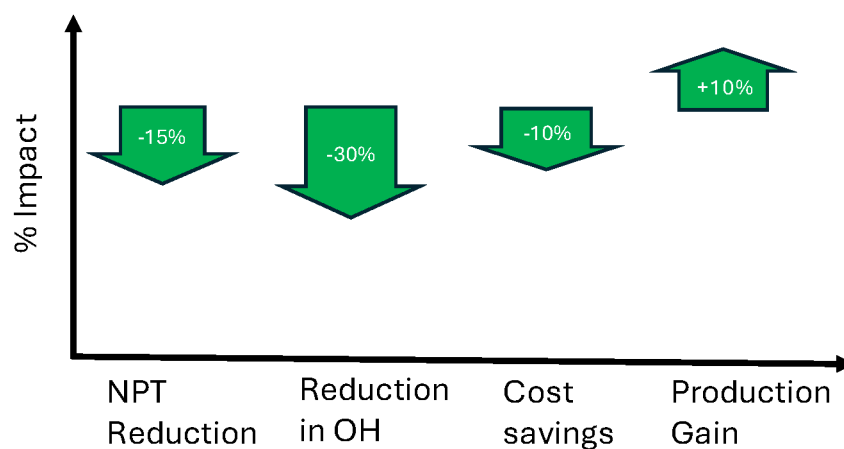


Figure 9—Realized benefits by implementation of operational excellence

The strong collaboration and common understanding across different disciplines, implementation of this approach had helped the well engineering and operations team to work-out an optimum drilling window with reduction of average time to drill reservoir section in new wells (improved by ~6 days). This has resulted in significant improvement in well delivery processes in reservoir

Novel/Additive Information

The integration of formation pressure from different sources and disciplines paved the way for a safer and robust well design and finding an optimal drilling window. Dynamic pore pressure testing (DPP) will further help to acquire the formation pressure data in areas of wider uncertainty and real-time ECD planning. Empirical correlations can be used from MPD data to enhance to formation pressure database and safe drilling practices.

Though single pressure communication might seem convenient and straight forward approach for users, however it is important to capture the full uncertainty range with clear communication to be able to achieve operational excellence in infill deliveries. The PPP workflow established by N. Kuwait Jurassic team by integrating MPD/drilling data in PPP process can be replicated in other assets with similar issues and environment to overcome the drilling challenges.

Acknowledgements

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Nomenclature

PPP	Pore Pressure Prediction
ECD	Equivalent Circulating Density
MPD	Managed Pressure Drilling
CG	Connection gas
DPP	Dynamic Pore Pressure Testing
XRD	X-Ray diffraction
MDT	Modular Formation Dynamics Tester XPT
HPHT	High Pressure High Temperature
FDP	Field Development Plan
NPT	Non Productive Time

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